

1. Identify the Problem Statement:
Insurance Charge Prediction
2. Approach : Machine Learning
-> The output column seems to be numerical so I have decided that the dataset belongs to the regression.
3. We have converted the String column values to numerical using nominal method.
4. We have tried out with 4 different models to get the better accuracy score.

Please look into the below mentioned models:

Model I: Multiple Linear Regression

The dataset shows that the multiple inputs and single output column, so that we chose the multiple linear regression.

R2_score : 0.7894790 < 0.99

Model II: Support Vector Machine

Based on the adjacent data points , it split the data and draw the linear and non-linear curves accordingly.

#	c	Gamma	#	LinearR2_score	#	PolyR2_score	#	RBFR2_Score	#	SigmoidR2_score
	100	scale		0.6359503		0.7508192		0.3906016		0.5275603
	100	auto		0.6359503		0.7508192		0.3906016		0.5275603
	500	scale		0.7651432		0.8593128		0.6964683		0.4906357
	500	auto		0.7651432		0.8593128		0.6964683		0.4906357
	1000	scale		0.7440908		0.8605846		0.8283522		0.1437755
	1000	auto		0.7440908		0.8605846		0.8283522		0.1437755
	2000	scale		0.7414238		0.8601812		0.8607356		-2.5840346
	2000	auto		0.7414238		0.8601812		0.8607356		-2.5840346
	8000	scale		0.7414203		0.85836		0.8769818		-47.8792585
	8000	auto		0.7414203		0.85836		0.8769818		-47.8792585

In this approach , the r2_score value is 0.8769818 with RBF kernel. Compared to all other kernel , the RBF r2score is better < 0.99, but somewhat near to 0.90.

Model III: Decision Tree Regressor

-> Basically this approach is like a tree structure .It will create the tree based on the decision.

Criterion ▾	Splitter ▾	Max_Features ▾	R2_Score ▾
squared_error	best	None	0.6905962
squared_error	random	None	0.6721923
squared_error	best	sqrt	0.6996484
squared_error	best	log2	0.6827724
squared_error	random	sqrt	0.6418461
squared_error	random	log2	0.6626151
friedman_mse	best	None	0.7227842
friedman_mse	random	None	0.7070838
friedman_mse	best	sqrt	0.77977409
friedman_mse	best	log2	0.7625934
friedman_mse	random	sqrt	0.5776251
friedman_mse	random	log2	0.5945083
absolute_error	best	None	0.68783001
absolute_error	random	None	0.77462708
absolute_error	best	sqrt	0.67035907
absolute_error	best	log2	0.7189626
absolute_error	random	sqrt	0.6404126
absolute_error	random	log2	0.7061935
poisson	best	None	0.7300693
poisson	random	None	0.68130708
poisson	best	sqrt	0.6750534
poisson	best	log2	0.6920852
poisson	random	sqrt	0.7035159
poisson	random	log2	0.7005231

This model has the R2_score value= 0.77977409.

Model IV: Random Forest

-> There is no pattern in selecting the model. It will create multiple models from that it will take the majority one and built a model.

n_estimators	crit	max_features	R2_score
100	squared_error	None	0.8575904
100	absolute_error	None	0.8498287
100	friedman_mse	None	0.8528867
100	poisson	None	0.8508557
100	squared_error	sqrt	0.8703312
100	absolute_error	sqrt	0.8698958
100	friedman_mse	sqrt	0.8672382
100	poisson	sqrt	0.8740538
100	squared_error	log2	0.8694804
100	absolute_error	log2	0.8693363
100	friedman_mse	log2	0.8710191
100	poisson	log2	0.8707574
500	squared_error	None	0.8544468
500	absolute_error	None	0.8552232
500	friedman_mse	None	0.85627706
500	poisson	None	0.8568616
500	squared_error	sqrt	0.8723356
500	absolute_error	sqrt	0.8723887
500	friedman_mse	sqrt	0.8710022
500	poisson	sqrt	0.8697465
500	squared_error	log2	0.8718865
500	absolute_error	log2	0.8710892
500	friedman_mse	log2	0.8712594
500	poisson	log2	0.8715859

This model has the R2_score = 0.8740538.

Conclusion:

This dataset has the good r2_score value in support vector regressor and Random Forest regressor. Finally These models got the r2_score value $0.87 < 0.99$ which is better when compared to the other models ie. Multiple Linear Regression and Decision Tree Regression which has the $0.78 < 0.99$ least R2_score.